

Problem: Let $v \in \mathbb{R}^n$ be a vector of unit norm ($\|v\|=1$)

Let $\alpha \in \mathbb{R}$ and $Q = I - \alpha vv^T$

For which values of $\alpha \in \mathbb{R}$ is Q an orthogonal matrix?

$$\begin{aligned} Q^T Q &= (I - \alpha vv^T)(I - \alpha vv^T) \\ &= I - \alpha vv^T - \alpha vv^T + \alpha^2 \underbrace{vv^T vv^T}_{v^T v = 1} \rightarrow \text{Diagram showing } v^T v = 1 \\ (I - \alpha vv^T)^T &= I^T - \alpha (v^T)^T v^T = I - \alpha vv^T \\ &\Rightarrow Q \text{ is symmetric (for any choice of } \alpha) \end{aligned}$$

$$\leadsto Q^T Q = I - 2\alpha vv^T + \alpha^2 vv^T = I + (\alpha^2 - 2\alpha) vv^T$$

$$\text{So } Q \text{ is orthogonal} \Leftrightarrow Q^T Q = I \Leftrightarrow \alpha^2 - 2\alpha = 0$$

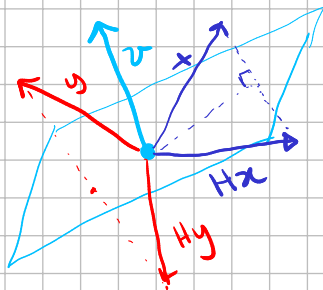
$$\Leftrightarrow \alpha = 0 \text{ or } \alpha = 2$$

$\downarrow$ not very interesting $\downarrow$:smiley:

Def: Given $v \in \mathbb{R}^n$, its associated Householder reflector is $H = I - \frac{2}{\|v\|^2} vv^T$

$\downarrow$
name of a person

$\downarrow$
reflection through the hyperplane perpendicular to v



Example: $v = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ $H = \begin{bmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix}$

$$H \cdot x = \begin{bmatrix} -x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Q: What is the computational cost of computing $H \cdot x$ for a vector $x \in \mathbb{R}^n$?

$$Hx = \left(I - \frac{2}{\|v\|^2} vv^T \right) x = x - \frac{2}{\|v\|^2} v \underbrace{v^T x}_{\text{compute this first (cost } O(n))}$$

1) compute $v^T x =: \beta$ $O(n)$

2) compute $\frac{2}{\|v\|^2} \cdot \beta =: \gamma$ $O(n)$

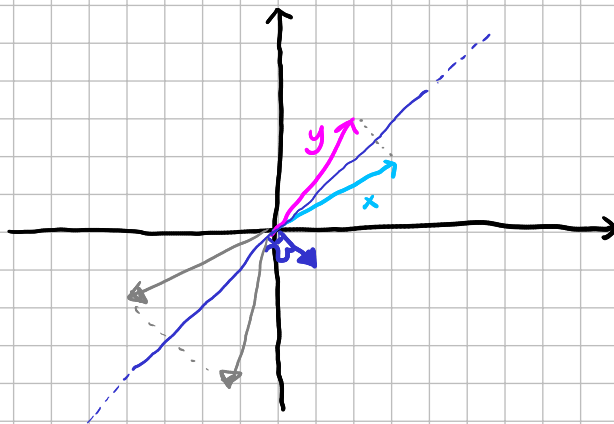
3) compute $x - \gamma v$ $O(n)$

Remark: $H^{-1} = H^T = H$

We can use Householder reflectors to transform a given vector $x \in \mathbb{R}^n$ into a given vector y using an orthogonal transformation.

$\hookrightarrow$ with $\|x\| = \|y\|$

Lemma: let $x, y \in \mathbb{R}^n$ with $\|x\| = \|y\|$ and let $v = x - y$
 Then the Householder reflector corresponding to v
 is such that $Hx = y$.



Rmk: if $Hx = y$ then $Hy = x$
 and we could have used
 the vector $y - x$ as well

As a consequence, given a vector $x \in \mathbb{R}^n$, we can find a
 Householder reflector H s.t. $Hx = s \cdot e$,
 where s is a scalar $\in \mathbb{R}$.

s can be $\|x\|$ or $-\|x\|$

option 1: $y = \|x\|e$, $\leadsto v = x - \|x\|e = x - se$
 option 2: $y = -\|x\|e$, $\leadsto v = x + \|x\|e = x + se$

[Matlab example]

$$x = \begin{bmatrix} 10^4 \\ 10^{-6} \\ 10^{-6} \\ 10^{-6} \end{bmatrix}$$

$$\|x\|^2 = 10^8 + 10^{-12} \cdot 3 < 10^{-16} \cdot 10^8$$

$$x = \begin{bmatrix} 1 \\ 10^{-10} \\ 10^{-10} \\ 10^{-10} \end{bmatrix}$$

$$\|x\|^2 = 1 + 3 \cdot 10^{-20} \stackrel{\text{matlab}}{\approx} 1$$

$$v = x - \|x\|e = \begin{bmatrix} 0 \\ 10^{-10} \\ 10^{-10} \\ 10^{-10} \end{bmatrix}$$

IMPORTANT ANNOUNCEMENT:

FORM GROUPS
 FOR THE PROJECTS!

deadline: Nov. 7

Theorem: For every matrix $A \in \mathbb{R}^{m \times n}$ there exist an orthogonal matrix $Q \in \mathbb{R}^{m \times m}$ and an upper triangular matrix R such that $A = QR \leftarrow QR$ decomposition of A

[$A = LU$ L lower triangular, U upper triangular]

$$A = Q \begin{matrix} \text{blue shaded triangle} \\ \text{blue dot} \end{matrix}$$

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Let's see a constructive proof!

step 1: $A = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$

choose a Householder reflector H_1 s.t.

$H_1 \cdot (\text{first column of } A) = \text{multiple of } e_1$

$$H_1 \cdot A = \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix} \rightarrow A_1$$

step 2: choose a Householder reflector $\tilde{H}_2$ s.t.

$\tilde{H}_2 \cdot (\text{first column of } A_1) = \text{multiple of } e_1$

$$\tilde{H}_2 \cdot \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{bmatrix} = \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{bmatrix}$$

let's define $H_2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \tilde{H}_2 & & \\ 0 & & & \\ 0 & & & \end{bmatrix} \rightsquigarrow \text{this is an orthogonal matrix!}$

$$H_2 \cdot (H_1 A) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \tilde{H}_2 & & \\ 0 & & & \\ 0 & & & \end{bmatrix} \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \\ 0 & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & * & * \end{bmatrix} \rightarrow A_2$$

step 3: choose a Householder reflector $\tilde{H}_3$ s.t.

$$\tilde{H}_3 A_2 = \tilde{H}_3 \cdot \begin{bmatrix} * & * \\ * & * \end{bmatrix} = \begin{bmatrix} * & * \\ 0 & * \end{bmatrix}$$

let's define $H_3 = \begin{bmatrix} I_2 & \tilde{H}_3 \\ 0 & \end{bmatrix} \rightarrow H_3 (H_2 H_1 A) = \begin{bmatrix} I & 0 & * & * & * & * \\ 0 & \tilde{H}_3 & 0 & * & * & * \\ & & 0 & 0 & * & * \\ & & 0 & 0 & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{bmatrix} \xrightarrow{R} A_2$

$$\leadsto \underbrace{H_3 H_2 H_1} A = R$$

this is the product of orthogonal matrices $\Rightarrow$ it's an orthogonal matrix!

$$A = (H_3 H_2 H_1)^T R = H_1^T H_2^T H_3^T R = \underbrace{H_1 H_2 H_3}_Q R$$

we constructed a QR factorization of A!

□

Q: What is the computational cost of constructing Q and R?

For a square matrix, it's $O(n^3)$ to form R

If we compute $Q = H_1 H_2 \dots H_{n-1}$ in the "naive way" then it costs $O(n^4)$, but we can actually compute Q in $O(n^3)$ because

$$H_j = \begin{bmatrix} I_{j-1} & 0 \\ 0 & \tilde{H}_j \end{bmatrix}$$

↑
Householder reflector

$$H_j \cdot X = \begin{bmatrix} I_{j-1} & 0 \\ 0 & \tilde{H}_j \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} \\ \tilde{H}_j x_{21} & \tilde{H}_j x_{22} \end{bmatrix}$$

$O(n^2 + n(n-j+1)) \leq O(n^2)$

$\Rightarrow$ the total cost of QR factorization is $O(n^3)$ for a square matrix and

[Ex: think why this is true] $\leftarrow$ it's $O(m^3 + m^2 n)$ for $m \times n$ matrices

Remark: relation with SVD: $\leftarrow$ orthogonal!

$$A = \underbrace{QR}_{\substack{\leadsto \\ = U_R \Sigma_R V_R^T}} = \underbrace{(QU_R)}_{\text{orthogonal!}} \Sigma_R V_R^T = \tilde{Q} \Sigma_R V_R^T$$

SVD of R

Q: If $A \in \mathbb{R}^{n \times n}$ is invertible, $b \in \mathbb{R}^n$, and a QR factorization of A is given to you: $A = QR$, what's the most efficient way of solving the linear system $Ax = b$?

$QRx = b \leadsto$ multiply by Q^T to the left, we get

$$\underbrace{Q^T Q} R x = Q^T b \rightarrow R x = \underbrace{Q^T b}$$

↳ cost $O(n^2)$

now we have an upper triangular linear system, which we can solve by backward substitution in $O(n^2)$.

The thin/economy QR factorization:

$$\begin{matrix} m \\ \left\{ \begin{array}{c} A \end{array} \right\} \\ n \end{matrix} = \begin{matrix} \left\{ \begin{array}{c} Q \end{array} \right\} \\ m \end{matrix} \cdot \begin{matrix} \left\{ \begin{array}{c} R \end{array} \right\} \\ n \end{matrix} \begin{matrix} n \\ \left\{ \begin{array}{c} Q_0 \end{array} \right\} \\ n \end{matrix} \begin{matrix} n \\ \left\{ \begin{array}{c} Q_c \end{array} \right\} \\ m-n \end{matrix} = \begin{matrix} \left\{ \begin{array}{c} R_0 \\ 0 \end{array} \right\} \\ m \end{matrix} = \begin{matrix} \left\{ \begin{array}{c} Q_0 \end{array} \right\} \\ m \end{matrix} \begin{matrix} \left\{ \begin{array}{c} R_0 \end{array} \right\} \\ n \end{matrix}$$

square $n \times n$ upper triang. matrix

In the Householder QR algorithm, we need only n steps to get to an upper triangular matrix $\leadsto$ corresponds to vectors $v_1, \dots, v_n$ used to build the Householder reflectors.

Trick: to compute Q_0 , it's sufficient to apply the matrices $H_1, H_2, \dots, H_n$ to $\begin{bmatrix} I_n \\ 0 \end{bmatrix}$

$$Q = H_1 H_2 \dots H_n$$

$$Q_0 = Q \cdot \begin{bmatrix} I_n \\ 0 \end{bmatrix} = \underbrace{H_1}_{m \times m} \underbrace{H_2}_{m \times m} \dots \underbrace{H_n}_{m \times m} \begin{bmatrix} I_n \\ 0 \end{bmatrix}^{m \times n}$$

$n=2$
After 1 step:

✖	✖
○	✖
○	✖
○	✖

Rmk: in the square $n \times n$ case, $n-1$ reflectors are enough. In the rectangular $n < m$ case, we need n reflectors.

Cost of computing a thin QR factorization of $A \in \mathbb{R}^{m \times n}$ with $m > n$ is $\approx 2mn^2$

Cost of computing a square QR fact. for $A \in \mathbb{R}^{n \times n}$ is $\approx \frac{4}{3}n^3$.